

19 Sept 2007

(b)

NUMBERS FROM SPACE

Ross Street
Macquarie University

Talk presented at the Open Day for Talented High School Students at the NSW Institute of Technology on 8 July, 1982.

There are many ways of extracting numbers from a space. I shall present some of these in this talk and explain some amazing connections between these numbers even though the methods of extracting them differ wildly.

1. Spaces. What do I mean by a space? Descartes taught us that a plane can be thought of as the set of pairs (x, y) of real numbers, and Euclidean 3-space as the set of triples (x, y, z) of real numbers. More generally, Euclidean n -space can be studied as the set $\mathbb{R}^n$ of n -tuples of real numbers.

For my purpose today a space can be taken to mean a subset of some Euclidean space, in fact, mainly bounded ones. I am not too interested in the position or size of the space in its ambient Euclidean space. The formula

$$x^2 + y^2 = 1$$

determines a subset of $\mathbb{R}^2$ called the circle (one circle being as good as another for our purposes). The sphere is the surface in $\mathbb{R}^3$ determined by

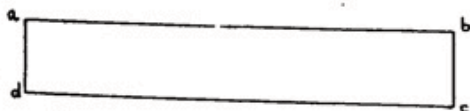
$$x^2 + y^2 + z^2 = 1.$$

Curves in $\mathbb{R}^3$ are generally defined by two equations in x, y, z . Surfaces in $\mathbb{R}^4$ are generally defined by two equations in the coordinates.

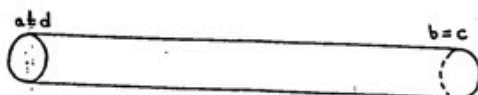
There are certain spaces called simplexes which play an important role in mathematics. A 0-simplex is a point, a 1-simplex is an interval, a 2-simplex is a triangle, a 3-simplex is a tetrahedron, and so on. To be specific the n -simplex Δ_n can be regarded as the subset of $\mathbb{R}^{n+1}$ consisting of those points whose first coordinate is no less than 0, whose second coordinate is no less than the first, ..., and 1 is no less than the last coordinate. For example,

$$\Delta_3 = \{(x, y, z) \mid 0 \leq x \leq y \leq z \leq 1\}.$$

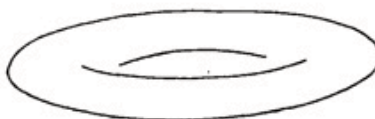
You will see that we can regard our surfaces as made of thin rubber sheeting and that stretching and bending will not change the numbers of interest. However, breaking or fusing the rubber leads to new surfaces. In fact, interesting surfaces can be built up from a rectangle by fusing the boundary in different ways.



Consider the above rectangle $abcd$. Pasting the line ab to the line dc , one obtains a cylinder.



Pasting the ends of this cylinder (corresponding to the original lines ad and bc) gives a "doughnut" or torus.

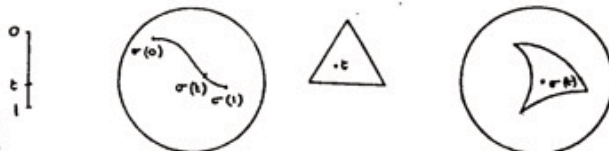


On the other hand, twisting the rectangle and gluing ad to cb gives a Mobius band.



In $\mathbb{R}^4$ it turns out to be possible to glue ab to dc and ad to cb without pushing the rectangle through itself; the result is the surface called the Klein bottle. Also in $\mathbb{R}^4$, ab can be glued to cd and ad to cb ; this gives the projective plane.

Spaces are mathematical entities much richer than mere sets. It is possible to speak of continuous functions between spaces. When we "draw a curve on a surface without lifting our pen" we are dealing with a continuous function from the interval (or 1-simplex) to the surface.



We can think of a continuous function σ from Δ_n to a space X as a bent version of the n -simplex living on (or in) X .

2. Euler characteristic.

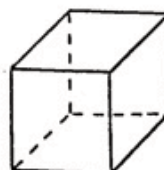
Descartes and Euler discovered that, for any polyhedron, if V is the number of vertices, E is the number of edges, and F is the number of faces, then

$$V - E + F = 2.$$

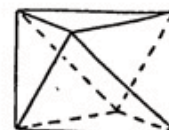
Using this, Euler was able to prove that the only regular polyhedra were the five already known to the Greeks (tetrahedron, cube, octahedron, icosahedron, and dodecahedron).



$$\begin{aligned} V &= 4 \\ E &= 6 \\ F &= 4 \end{aligned}$$



$$\begin{aligned} V &= 8 \\ E &= 12 \\ F &= 6 \end{aligned}$$



$$\begin{aligned} V &= 6 \\ E &= 12 \\ F &= 8 \end{aligned}$$

These topics are well treated in the book:

Courant, Robbins: "What is mathematics?" (Oxford U. Press).

The numbers V, E, F do not change if we regard the polyhedron as made of a rubber sheet and start stretching and bending. So we may as well regard all polyhedra as drawn on a sphere. Drawing in diagonals in polygonal faces, we do not change $V - E + F$; so we may as well consider only polyhedra with all faces triangular. To give such a polyhedron is called a triangulation of the sphere. More generally, a triangulation of a space X amounts to the giving of a finite number of continuous functions from simplexes such that these bent simplexes on the space X fill up all of X and only meet along faces. (We use the term face to cover "end-points of intervals", "sides of triangles", "faces of tetrahedra", and so on.)

For a particular triangulation of a space X , we can let Q_n be the number of n -dimensional faces in the triangulation. So Q_0 is V , the number of vertices; Q_1 is E , the number of edges; Q_2 is F , the number of 2-dimensional faces; and so on. For a surface it stops at Q_2 . Then we can consider the number

$$\chi(X) = Q_0 - Q_1 + Q_2 - Q_3 + \dots$$

For X taken to be the sphere, Euler's result was that, although the numbers Q_n depend on the triangulation, the number $\chi(X)$ does not. This viewpoint on the Descartes-Euler formula is due to Poincaré. It is difficult to prove, yet true, that $\chi(X)$ does not depend on the triangulation for any X which has a triangulation.

The number $\chi(X)$ is called the Euler characteristic of the space X .

It is amazing how many properties of a space depend purely on its Euler characteristic.

By drawing polyhedra on a torus one soon convinces oneself that the Euler characteristic of a torus is 0. The Euler characteristic of projective space is 1, and of the Klein bottle is -1.

3. Genus.

The genus $g(X)$ of a surface is the largest number of non-intersecting simple closed curves that can be drawn on the surface such that, if one were to cut around these curves and shake, the surface would still hang together.

A sphere has genus 0, a torus has genus 1. Siamese doughnuts have genus 2. The sphere with two handles has genus 2.



For two-sided surfaces (such as spheres with handles), the Euler characteristic and genus are related by the formula:

$$\chi(X) = 2 - 2g(X).$$

For one-sided surfaces (such as the Klein bottle), we have:

$$\chi(X) = 2 - g(X).$$

4. Betti numbers.

Before discussing Betti numbers I need a little terminology from set theory. If A, B are subsets of a set S , let $A \Delta B$ denote the subset of S consisting of those elements which are in A or B but not in both. I write 0 for the empty set.

Notice:

$$\begin{aligned} (A \Delta B) \Delta C &= A \Delta (B \Delta C) \\ A \Delta B &= B \Delta A \\ A \Delta A &= 0 \\ A \Delta 0 &= A. \end{aligned}$$

Now suppose X is a space. Let X_n denote the set of all continuous functions

$$\sigma: \Delta_n \rightarrow X.$$

So X_0 is the set of points of X ; X_1 is the set of continuous paths on X ; X_2 is the set of curly triangles in X .

From each subset A of X_n , I can define a subset ∂A of X_{n+1} as follows:

$$\partial A = \{ \sigma \in X_{n+1} \mid \text{an odd number of faces of } \sigma \text{ are in } A \}.$$

For example, if A is a set of points of X then ∂A is the set of all paths with exactly one end in A .

Theorem $\partial \partial A = 0$.

Proof. Each face of a face is in precisely two faces. (For example, each edge of a tetrahedron is in precisely two faces of the tetrahedron.) A little thought shows this gives the result. $\square$

A subset of X_n of the form ∂B for some subset B of X_{n-1} is called an n -coboundary. Note that 0 is a coboundary.

A subset A of X_n is called an n -cocycle when $\partial A = 0$. The above Theorem can be restated: every coboundary is a cocycle.

The Betti numbers $\beta_n(X)$ of X measure the extent to which there are cocycles which are not coboundaries. More precisely, $\beta_n(X)$ is the maximum number m of n -cocycles $A_1, A_2, \dots, A_m$ such that

$$A_1 \Delta A_2 \Delta \dots \Delta A_m$$

is not a coboundary.

Amazingly, although the number of subsets of X_n is horribly infinite and so too is the number of n -cocycles, we do have the following result.

Theorem. If X is a space which admits a triangulation then each $\beta_n(X)$ is finite and:

$$\chi(X) = \beta_0(X) - \beta_1(X) + \beta_2(X) - \beta_3(X) + \dots \quad \square$$

Let us look a little closer at the meaning of $\beta_0(X)$ and $\beta_1(X)$.

The only 0-coboundary is 0. A 0-cocycle is a set P of points of X such that any path with one end in P has the other end in P too. If we say that two points of X are in the same connected component when there is a path in X with these two points as ends, then we see that each connected component of X gives a 0-cocycle. These considerations lead to the result that $\beta_0(X)$ is the number of connected components of X .

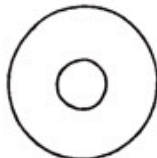
If P is a set of points in X , ∂P is the set of paths in X with exactly one end in P . A set A of paths in X is a 1-cocycle when, for any triangle in X , if one side is 'in' A then precisely one other side is in A .

A subset X of Euclidean space is said to be star-shaped when there is a point p in X such that, for all points x in X , the line segment joining p to x is entirely in X .

Theorem. If X is star-shaped then $\beta_n(X) = 0$ for $n > 0$.

Proof. I shall prove the case $n = 1$, although the argument generalises. Let A be a 1-cocycle. We must find a set P of points such that $A = \partial P$. The set P is the set of points x such that the line joining x to p is a path in A . Now take any path σ in X . We obtain a triangle τ in X by joining up the points on σ to p . Since A is a 1-cocycle, σ is in A precisely when one of the other sides of τ is in A ; but this occurs precisely when exactly one of the ends of σ is in P ; and this occurs precisely when σ is ∂P . So A and ∂P have the same elements. So $A = \partial P$ as required. $\square$

The argument in the proof above does not work when X has a hole in it, such as the annulus below.



In fact, $\beta_1(X)$ is the number of holes in X .

For any surface, $\beta_n(X) = 0$ for $n > 2$.

For the sphere $\beta_0 = 1$, $\beta_1 = 0$, $\beta_2 = 1$.

For the torus, $\beta_0 = 1$, $\beta_1 = 2$, $\beta_2 = 1$.

So indeed we do have $\chi = \beta_0 - \beta_1 + \beta_2$ in these cases.

5. Chromatic number.

For a polyhedron drawn on a surface, what is the smallest number of colours needed to colour all faces such that faces sharing a common edge have different colours? This is the chromatic number of the surface. It is easy to see that the case of a map in the plane is the same as a polyhedron on a sphere (by declaring one face 'the ocean'). However it turned out that this was the hardest case to solve. It was known for a long time that at most 5 colours were needed, but no case was known on the sphere which needed more than 4. This was the famous four colour problem, which has recently been solved.

The first surface for which the problem was solved was the torus where the chromatic number is 7. The following was proved by Heawood in 1890.

Theorem. Let χ be the Euler characteristic of a (compact connected) surface X and let N_X be the largest integer less than or equal to

$$(7 + \sqrt{49 - 24\chi})/2$$

If $\chi \leq 0$ then the chromatic number of X is $\leq N_X$. $\square$

In 1934 Franklin found that only 6 colours are needed for the Klein bottle whereas $N_X = 7$.

χ	2	1	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10
N_X	4	6	7	7	8	9	9	10	10	10	11	11	12

The chromatic number of the projective plane was known to be 6.

Ringel and Youngs (1968) proved that, except for the Klein bottle, every (compact connected) surface of characteristic $\chi \leq 0$ has chromatic number N_X .

Four Colour Theorem. (Appel-Haken 1976) The chromatic number of the sphere is 4.

The proof of this result made extensive use of the computer once general methods had reduced the possible exceptional polyhedra to a large finite number of complicated polyhedra.

6. Curvature.

Suppose X is a nice surface in $\mathbb{R}^3$. One can construct a continuous function γ from X to the sphere with $\gamma(p)$ equal to the unit normal to the surface X at the point p . (A normal at p is a line perpendicular to the tangent plane at p ; and unit means of unit length; and unit vectors can be identified with points on the sphere

$$x^2 + y^2 + z^2 = 1.)$$

The Gaussian curvature of X at p is defined by;

$$K(p) = \lim_{A \rightarrow P} \frac{\text{area } \gamma(A)}{\text{area } A}$$

where A is a small neighbourhood of p on X and $\gamma(A)$ denotes its image under the function γ . It is possible for $K(p)$ to be negative.

Gauss-Bonnet Theorem. If X is made of a thin material whose density at point p is $K(p)$ then the weight of X is $2\pi\chi(X)$. $\square$

That Euler characteristic keeps cropping up!